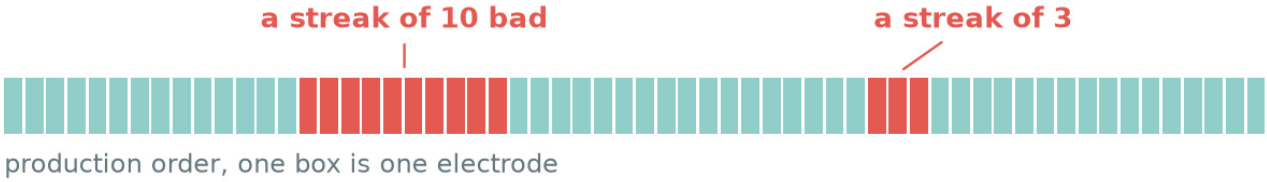
How we decide how often to test

1. Faults arrive in streaks

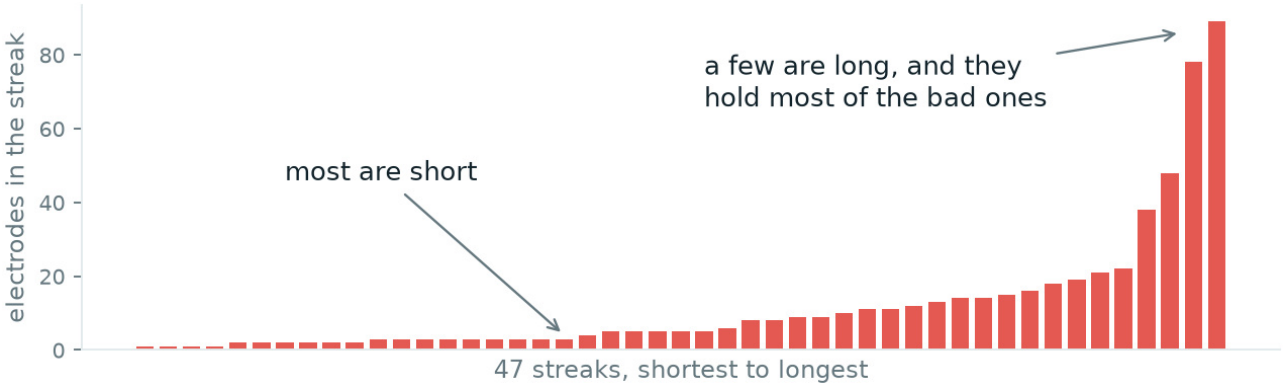
A bad electrode is almost never on its own. Something in the process drifts, the line makes bad electrodes for a while, somebody fixes it, and the line goes back to normal. So production looks like this:



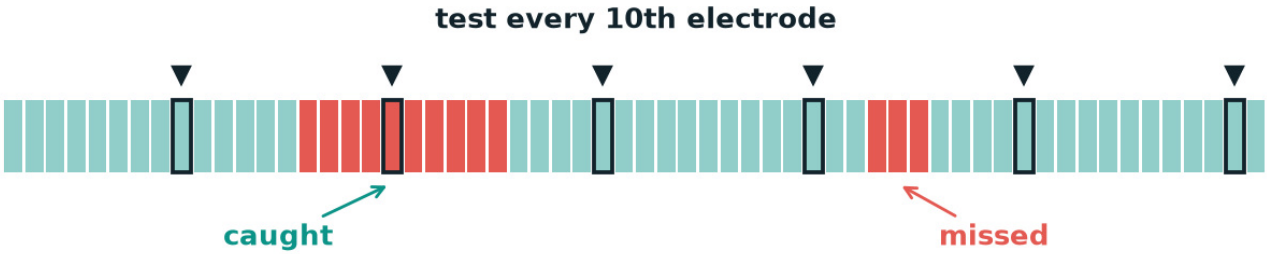
That single fact is what makes sampling work at all. If bad electrodes were scattered at random, testing one in twenty would find one in twenty of them and nothing more. Because they come in streaks, one test can expose a whole stretch of bad production.

2. Streak length statistics

Vat 28, every electrode in the archive: **47 streaks**, made of 561 bad electrodes.



3. Chance of catching a streak



A streak is caught only if a test happens to land inside it, so the longer it runs, the harder it is to miss. A shorter one is caught only in proportion to its length, so half as long as the interval means caught about half the time. Short streaks are the ones that slip through.

4. Target catch rate

15 % production fault rate, with bad streaks averaging 12 electrodes

Finding this share of bad electrodes	Test one in every	Good electrodes destroyed
70 %	**26**	3.3 %
80 %	**18**	4.7 %
90 %	**10**	8.5 %
95 %	**5**	17.0 %

3 % production fault rate, with bad streaks averaging 3 electrodes

Finding this share of bad electrodes	Test one in every	Good electrodes destroyed
70 %	**5**	19.4 %
80 %	**3**	32.3 %
90 %	**2**	48.5 %
95 %	**2**	48.5 %

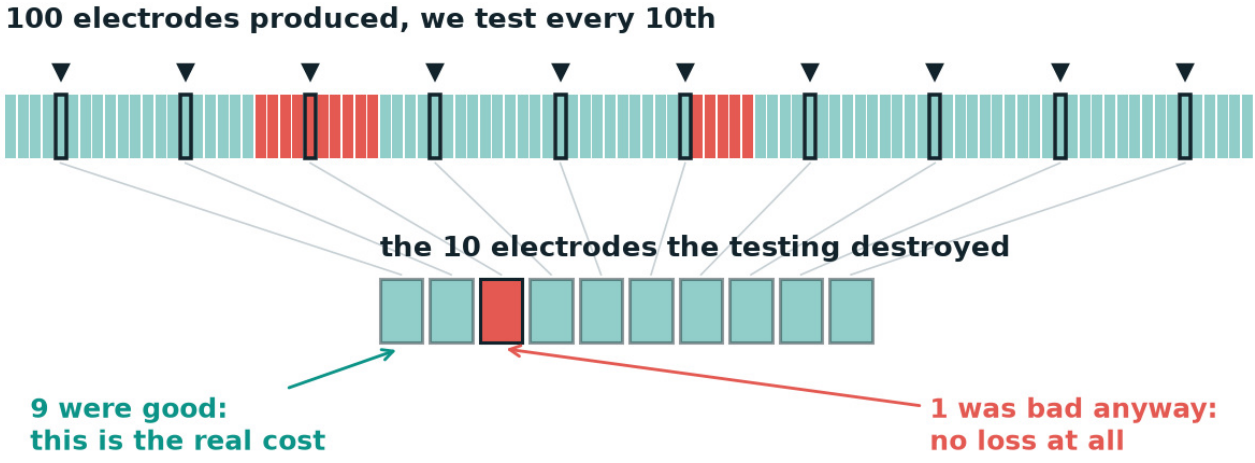
Setting a target catch rate for bad electrodes is not the right way of thinking. Catching 90% of bad electrodes is completely different when the share of bad electrodes is low or high. When the fault rate is low, catching most electrodes becomes really expensive. Therefore, when setting a sampling rate we should choose a number that does not destroy too many good electrodes.

We should come up with a formula that takes into account the cost of testing.

5. Cost and benefit of testing

Every time we test an electrode, we destroy it. That is a loss for our company. But is it really? It is only a real loss if the electrode is good. If it is bad, destroying it has cost us nothing. Everything below is counted per 100 electrodes produced, so that the cost and the benefit can be compared directly. Out of every 100 electrodes we make, this is how many good ones the testing destroys:

$$\text{good destroyed} = \frac{100 \times (1 - \text{fault rate})}{\text{sampling interval}}$$



On the other hand, what we gain by testing is not only the bad electrodes that have been tested, but the

entire streaks of bad electrodes that we do not sell. Out of the same 100 electrodes, this is how many bad ones the testing keeps from being sold:

$$\text{bad stopped} = 100 \times \text{fault rate} \times \text{catch rate}$$

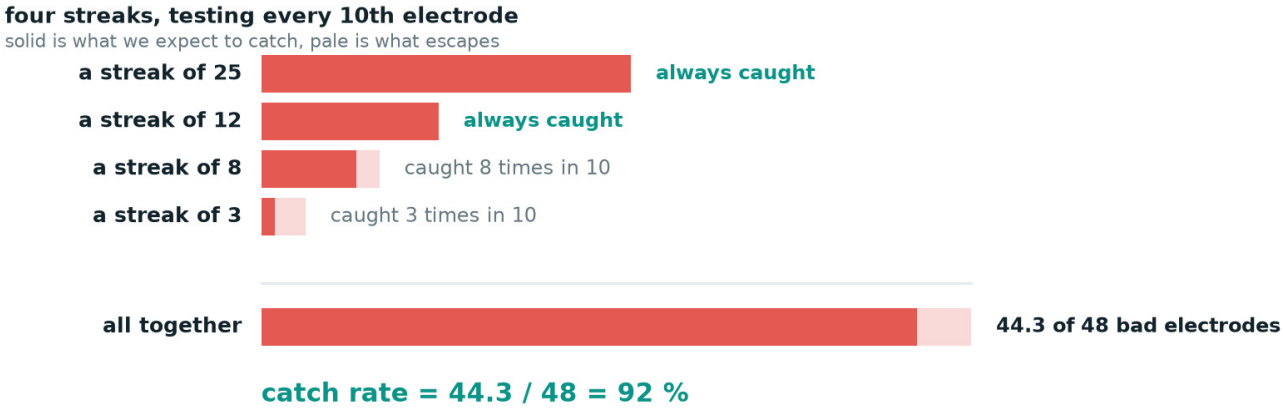
, where

$$\text{catch rate(sampling interval)} = \frac{\sum L_i \times \min(1,\ L_i/\text{sampling interval})}{\sum L_i}$$

, where

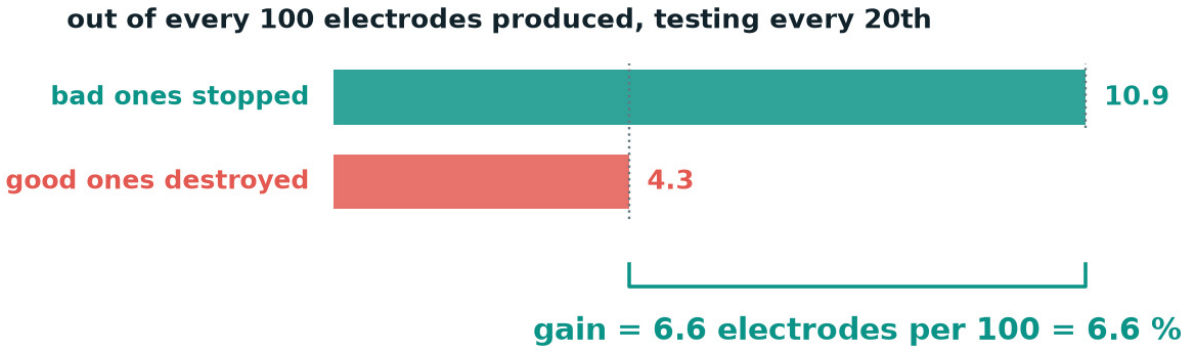
$$L_i = \text{length of bad streak } i$$
$$\sum L_i = \text{number of bad electrodes produced}$$

A streak is caught in proportion to how long it is, and once it is longer than the sampling interval it is always caught. Each streak then counts for as many electrodes as it holds, so a long one weighs more than a short one.

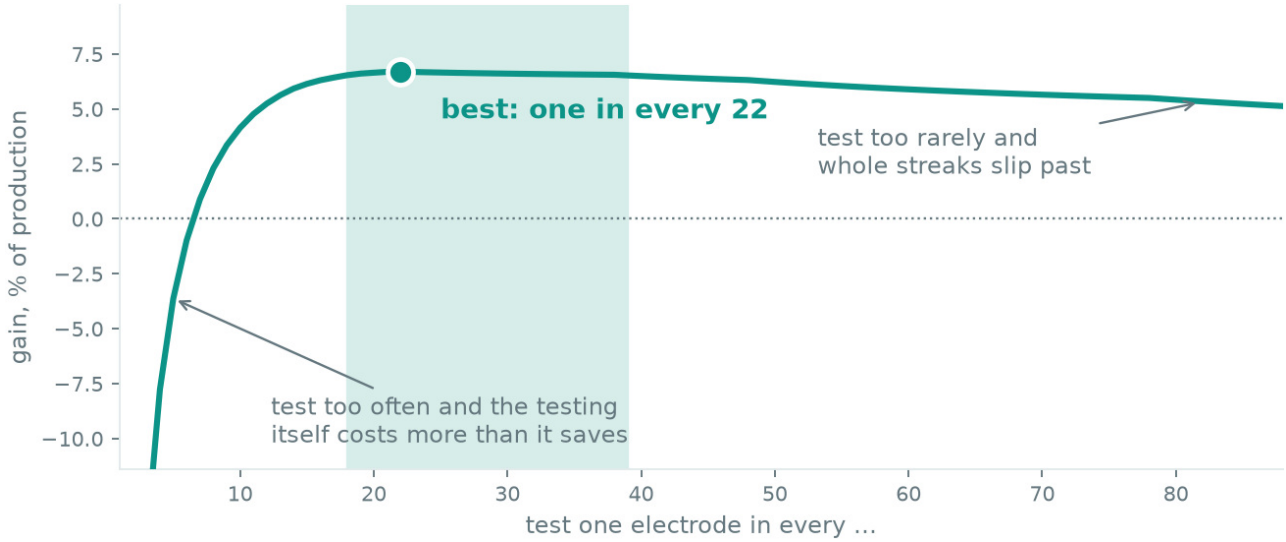


6. Gain

$$\text{gain} = \text{bad stopped} - \text{good destroyed}$$



Gain means the net benefit of testing, measured in percentage. If we stop more bad electrodes from reaching customers than we destroy in the process, we have a positive gain. The algorithm simply chooses the sampling rate that maximizes this gain. The plot of the gain as a function of the sampling interval is shown below.



The **gain** tries to balance between too frequent and too infrequent testing. Too infrequent testing allows bad electrodes to reach customers, while too frequent testing destroys good electrodes unnecessarily.

7. What the model assumes

The gain counts every bad electrode we stop as worth a whole electrode, which is the same as assuming that a bad electrode reaching a customer is worth nothing at all. That is the one place in the whole calculation that rests on judgement rather than on something we measured.

Putting a more accurate number on it is challenging. We can measure how a bad electrode performs and say that it runs at, for example, 70 % of the efficiency of a good one. That does not make it worth 70 % of a good one. The customer paid for a full electrode and expects a full electrode, so the loss is not simply the missing 30 %. It is that shortfall plus whatever it costs us in claims, replacements and standing with that customer, and none of that can be priced with any confidence. Treating a shipped bad electrode as worth nothing keeps the assumption in one visible place rather than hiding a guess inside the arithmetic. If a defensible figure for what a shipped fault really costs ever appears, it goes into the same formula and the recommended interval moves with it.